Week 3: The Forward Diffusion Process

Theme: Learn how to systematically destroy an image with noise (Useful)
Time Commitment: 8–12 hours
Suggested Split: 4h reading/video (math-heavy week), 5h coding, 1h review

Learning Objectives

By the end of this week, you should be able to:

- Explain the forward diffusion process and why it's a Markov chain
- Implement linear and cosine noise schedules
- Use the closed-form $q(x_t|x_0)$ to sample x_t in one step
- Build the dataset pipeline that produces (x_t, t, ϵ) training tuples

Primary Resources (Must-Do)

[BLOG/DOCS] Lilian Weng: "What are Diffusion Models?"

Type: Blog Post | **Time:** 60–90 min | **Difficulty:** Intermediate-Advanced
URL: https://lilianweng.github.io/posts/2021-07-11-diffusion-models/

Why this resource: The single best resource on diffusion models in existence. Read this twice if needed. The math notation here is the standard everyone uses.

Focus on: Sections 1 (What are Diffusion Models) and 2 (Forward diffusion process). You'll come back for the rest in Week 4.

[PAPER] DDPM Paper: "Denoising Diffusion Probabilistic Models" (Ho et al., 2020)

Type: Research Paper | **Time:** 90 min | **Difficulty:** Advanced
URL: https://arxiv.org/pdf/2006.11239

Why this resource: The paper that started the modern diffusion era. Don't be intimidated — with Lilian Weng's blog as a companion, this becomes approachable.

Focus on: Sections 1 and 2 (Background) this week. The notation in the paper is the same as Lilian Weng's — they reinforce each other.

[VIDEO] Outlier: "Diffusion Models | Paper Explanation | Math Explained"

Type: YouTube Video | **Time:** 33 min | **Difficulty:** Intermediate
URL: https://www.youtube.com/watch?v=HoKDTa5jHvg

Why this resource: The clearest visual walkthrough of the forward diffusion math. Specifically built for people learning diffusion for the first time.

Focus on: The reparameterization trick portion — this is what enables the closed-form sampling.

[VIDEO] Ari Seff: "What are Diffusion Models?"

Type: YouTube Video | **Time:** 16 min | **Difficulty:** Beginner-Intermediate
URL: https://www.youtube.com/watch?v=fbLgFrlTnGU

Why this resource: Beautiful 3D-animated explanation of the diffusion process. Watch this first if the math feels overwhelming — it builds intuition before formalism.

This Week's Deliverable

Implement the forward diffusion process. Code must include:

- A `NoiseScheduler` class supporting both linear and cosine schedules
- Pre-computed buffers for β_t, α_t, $\bar{\alpha}_t$, $\sqrt{\bar{\alpha}_t}$, $\sqrt{1-\bar{\alpha}_t}$
- A method `add_noise(x_0, t)` that returns x_t in one step (using the reparameterization trick)
- A visualization script: take one image and show it noised at $t = 0, 100, 250, 500, 750, 999$ in a single figure
- Unit tests verifying that x_T is approximately $\mathcal{N}(0, I)$ (check mean ≈ 0, std ≈ 1)

No model training this week. Just the data pipeline. This is a quieter coding week — use the saved time to absorb the math deeply.

Key Concepts Checklist

You should be able to confidently answer all of the following:

- Why can we sample x_t directly from x_0 in one step instead of iterating t times?
- What's the difference between β_t, α_t, and $\bar{\alpha}_t$?
- Why is the cosine schedule generally better than the linear schedule?
- What does the reparameterization trick do and why do we need it?
- Why must image data be normalized to $[-1, 1]$ before adding noise?
- As $t \to T$, what does x_t approach? Why does that matter for sampling?

Stretch Resources (Optional)

- **Improved DDPM paper** (Nichol & Dhariwal, 2021): https://arxiv.org/pdf/2102.09672 — introduces the cosine schedule
- **Calvin Luo: "Understanding Diffusion Models: A Unified Perspective"**: https://arxiv.org/pdf/2208.11970 — the most rigorous mathematical treatment
- Plot the signal-to-noise ratio (SNR) curve $\bar{\alpha}_t/(1-\bar{\alpha}_t)$ for both linear and cosine schedules
- Implement and visualize a quadratic and a sigmoid schedule

Debugging & Reference

- **Common pitfall:** Confusing α_t and $\bar{\alpha}_t$ leads to wildly wrong noising. Print both arrays and compare.
- **Numerical stability:** Use `torch.cumprod` carefully near $t = T$ where values approach zero.
- **Reference notation guide:** Pin a one-page summary of all the diffusion symbols above your desk.